

# Using RL and Offline Datasets for Fine-tuning LLMs

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## Abstract

*Fine-tuning Large Language Models (LLMs) using offline datasets introduces challenges when these datasets are noisy or sub-optimal. In this work, we investigate the impact of such sub-optimal supervision by structurally corrupting the GSM8K dataset and explore reinforcement learning (RL) techniques to mitigate its effects. We propose a hybrid-RL-based approach that integrates both offline and online learning, enabling the model to start from sub-optimal policies and iteratively improve through reward-driven updates. This is compared against a standard Supervised Fine-Tuning (SFT) baseline, and a Proximal Policy Optimization (PPO) baseline. While SFT performance degraded under increasing corruption, standard PPO was partially effective in recovering performance. However, due to GPU memory limitations, complete PPO and hybrid PPO runs could not be executed. Despite this, initial PPO updates demonstrated improved accuracy over SFT under 50% corruption, and Hybrid PPO showed signs of diverging from suboptimal behaviors.*

## 1. Introduction

The development of modern Large Language Models (LLMs) follows a two-stage life cycle, beginning with a broad pre-training phase before specializing through a more focused post-training process. During pre-training, the model is trained to learn general-purpose language representations and reasoning capabilities. However, this broad training is often not sufficient for acceptable levels of performance on downstream tasks or aligning with human expectations and preferences. Post-training techniques are the methodologies employed after a model has undergone its initial, general pre-training, with the goal of refining and adapting the model for specific tasks or user requirements [3].

Within post-training, two dominant paradigms have emerged to adapt these general models: Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL). One primary method is SFT, which enhances the reasoning capa-

bility of LLMs by fine-tuning them with Chain-of-Thought (CoT) annotations [3]. Another key approach is RL, which provides a formalism for learning control policies directly from experience [2]. Many modern approaches combine these methods, often using an SFT or imitation learning phase as a “warm-up” before further refinement with an RL algorithm like Proximal Policy Optimization (PPO) [3, 5].

Despite their power, both SFT and RL present a fundamental trade-off between performance, data cost, and computational resources. High-quality SFT often depends on imitation learning from human demonstrations, but collecting this data is expensive and not scalable [5]. For example, collecting 80,000 demonstrations for a single navigation task took approximately 2,894 hours of human teleoperation and cost \$50,000 [5]. While RL is more scalable, the classic formulation requires a lengthy active exploration process for each behavior, making it difficult to apply in many real-world settings [4]. Furthermore, it is exceptionally difficult to train a policy with potentially sub-optimal offline data and then successfully improve it further with online RL [4].

To navigate this trade-off, our work is predicated on the hypothesis that sub-optimal, or imperfect, datasets contain valuable signals for learning. We believe that allowing RL algorithms to effectively use previously collected data can make applications more practical [4]. This prior data, even if sub-optimal, can provide a starting point that mitigates challenges related to exploration and sample complexity by illustrating potentially useful transitions [4]. Therefore, we propose a novel hybrid RL algorithm designed to efficiently bridge the gap between learning from existing offline data and refining performance with online interaction. The aim is to develop a simple yet effective algorithm that can leverage large amounts of offline data for pre-training and then quickly perform online fine-tuning of RL policies [4]. This approach is designed to be effective for pre-training from off-policy datasets while also being well-suited to continuous improvement with online data collection [4].

To validate our approach, we conduct experiments evaluating the accuracy of our proposed algorithm against established baselines on the GSM8k dataset, which is a standard

mathematical reasoning benchmark, and consists of 8500 grade school-level math word problems, each paired with a detailed CoT solution and a final numeric answer [1, 3]. We create sub-optimal versions of the GSM8k dataset by structurally corrupting both the CoT and the final answer, and compare the performance of our algorithm (hybrid-RL) against a standard SFT baseline and a PPO baseline. We observe that the performance of the SFT baseline degrades with corruption in the dataset; however, using RL-based fine-tuning with a sub-optimal starting point enhances performance.

## 2. Related Work

A common first step in adapting Large Language Models (LLMs) is Supervised Fine-Tuning (SFT), which relies on high-quality demonstration data to establish a baseline of performance. SFT is a post-training technique that adapts pre-trained models to specific tasks by leveraging task-specific labeled datasets [2]. This process often serves as a "warm-up" stage in a larger training pipeline, equipping the model with the fundamental ability to solve problems before more advanced training [3]. This initial training typically takes the form of imitation learning, specifically behavior cloning, where the model learns to mimic actions from a dataset of human demonstrations [5].

Following SFT, Reinforcement Learning (RL) is often used to further refine model behavior by optimizing policies based on feedback signals. A prominent method is Reinforcement Learning from Human Feedback (RLHF), where human preference data is used to train a reward model [2]. An RL algorithm, such as Proximal Policy Optimization (PPO), is then used to iteratively adjust the policy to maximize the rewards from this model [2, 3]. This is generally accomplished using an actor-critic setup, which alternates between policy evaluation (estimating the value of actions) and policy improvement (updating the model’s policy based on those value estimates) [4].

However, a significant bottleneck for both SFT and RL-based methods is their dependency on the high cost and limited scalability of demonstration datasets. Fine-tuning often relies on large datasets of human demonstrations, which are expensive and time-consuming to collect, thus limiting their scalability [5]. In the same way that computer vision and NLP models benefit from large-scale pre-training datasets, practical applications of RL need to be able to incorporate large amounts of prior data to become more practical [4]. This reliance on costly human-labeled data significantly limits the scalability and efficiency of many alignment techniques [2].

To address the data dependency problem, offline RL methods have been developed to utilize static, pre-collected datasets. Offline RL methods aim to address the data issue by leveraging static datasets, often by constraining the

model to avoid deviating from the data distribution [2, 4]. While these methods perform well in purely offline settings, they struggle with online fine-tuning because they tend to be overly conservative [4]. This conservatism, designed to prevent error accumulation on the static dataset, severely limits the model’s ability to improve when new online data is introduced [4].

Our work seeks to bridge the gap between offline pre-training and online fine-tuning with a hybrid approach that effectively integrates sub-optimal data. Our proposed method differs from a standard SFT+PPO pipeline, where SFT is only used as an initial warm-up [3, 5]. Instead, we aim to integrate the learning signal from the offline, sub-optimal dataset directly into the online RL optimization process [4]. This allows our method to avoid the excessive conservatism of pure offline RL algorithms while still effectively leveraging prior data to accelerate and improve online learning [4, 5].

## 3. Methodology

We investigate the performance of different training paradigms on a corrupted version of the GSM8K dataset. We deliberately introduce structural noise into the dataset to simulate imperfect supervision, creating a suboptimal dataset. For our experiments, we start with a pretrained Galactica Small model (125M parameters)

To address this task, we explore three training strategies: (1) **Supervised Fine-Tuning (SFT)**, where the model is trained using behavior cloning on the corrupted dataset; (2) **standard Proximal Policy Optimization (PPO)**, which optimizes the language model using reinforcement learning with rewards based on answer correctness; and (3) a novel **hybrid PPO** approach that combines offline learning from the corrupted dataset with online policy updates guided by reward feedback. The following sections describe our training setup, reward design, and implementation details for each approach.

### 3.1. Corrupting Training Data

To simulate noisy supervision in the GSM8K training set, we introduce controlled structured corruption by replacing a fraction of the training samples’ final answers and corresponding CoT explanations with those from randomly selected different examples. The corruption ratio determines what percentage of the training set is altered. This perturbation preserves the original question but replaces the final answer and reasoning with mismatched content, allowing us to study the model’s robustness to inconsistencies in supervision signals. A corruption ratio of 0 corresponds to an unaltered, fully optimal dataset, while a ratio of 1 represents a fully corrupted and sub-optimal training set. In our experiments, we compare model performance under two conditions: no corruption (ratio = 0) and partial corruption (ratio

= 0.5).

### 3.2. Supervised Fine-Tuning

For supervised fine-tuning (SFT), the model is trained to maximize the likelihood of generating the gold chain-of-thought (CoT) tokens provided in the dataset. This is formulated as a standard language modeling objective that minimizes the negative log-likelihood (NLL) loss:

$$\mathcal{L}_{\text{SFT}} = - \sum_{t=1}^T \log \pi_{\theta}(a_t \mid a_{<t}, x) \quad (1)$$

where  $x$  is the input question,  $a_{1:T}$  are the gold CoT tokens, and  $\pi_{\theta}$  is the model parameterized by  $\theta$ .

This objective encourages the model to imitate the trajectories in the dataset. However, if the dataset is suboptimal or contains noisy reasoning steps, the model may overfit to these flawed demonstrations, ultimately degrading performance during inference.

### 3.3. Baseline SFT+PPO Pipeline

The baseline process begins with a "warm-up" stage where the policy is fine-tuned using Supervised Fine-Tuning (SFT) on a given dataset. This initial training is a form of imitation learning that uses behavior cloning to pre-train the policy on expert demonstrations. The objective is to learn a policy,  $\pi_{\theta_{BC}}$ , that minimizes the negative log-likelihood of the actions in the demonstration dataset. Following this SFT phase, the model is further refined using an on-policy reinforcement learning algorithm, specifically Proximal Policy Optimization (PPO). The PPO algorithm works to maximize a clipped objective function based on an advantage estimate, which compares the return of a specific action against the average value of the state. The novel rollouts allows the model to explore answers beyond the dataset.

The PPO algorithm optimizes the following clipped surrogate objective: The clipped surrogate objective used in PPO is defined as:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2)$$

where

$$r_t(\theta) = \frac{\pi_{\theta}(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \quad (3)$$

and  $\hat{A}_t$  is the estimated advantage at timestep  $t$ .

is the probability ratio between the new and old policies, and  $\hat{A}_t$  is the estimated advantage at time step  $t$ . The clipping function prevents the new policy from deviating too far from the old one by restricting  $r_t(\theta)$  to the interval  $[1 - \epsilon, 1 + \epsilon]$ .

### 3.4. Reward Function

The reward function is designed to encourage accurate and interpretable answer extraction while constraining the learned policy to remain close to the initial warm-up policy.

We define the total reward at each timestep as a combination of task reward and a KL penalty:

$$r_t = r_t^{\text{task}} - \beta \cdot \text{KL}(\pi_{\theta}(\cdot \mid s_t) \parallel \pi_{\text{ref}}(\cdot \mid s_t)) \quad (4)$$

where  $r_t^{\text{task}}$  is defined as:

- 1.0 if the extracted answer is exactly correct,
- 0.1 if the extracted answer is a number but incorrect, and
- 0.0 otherwise.

The KL divergence term penalizes deviation from the warm-up (reference) policy  $\pi_{\text{ref}}$ , with  $\beta$  controlling the strength of this regularization. This encourages stable learning and prevents the policy from drifting too far from the behavior encoded in the initial dataset.

### 3.5. Hybrid PPO

Our proposed algorithm extends the standard PPO framework by incorporating offline trajectories collected using a warm-start policy. This policy—trained via supervised fine-tuning—serves as the reference policy that generated the sub-optimal dataset. To leverage these trajectories for on-policy updates, we use an Importance Sampling-based Generalized Advantage Estimation (IS-GAE) approach, which reweights the advantages computed from the offline data.

The IS-GAE estimate of the advantage at time step  $t$  is given by:

$$\hat{A}_t^{\text{IS-GAE}} = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \left( \prod_{k=0}^{l-1} r_{t+k} \right) \delta_{t+l} \quad (5)$$

where

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \quad (6)$$

and

$$r_t = \frac{\pi_{\theta}(a_t \mid s_t)}{\pi_{\text{ref}}(a_t \mid s_t)} \quad (7)$$

is the importance weight ratio between the current policy  $\pi_{\theta}$  and the reference (offline) policy  $\pi_{\text{ref}}$ .  $\gamma$  is the discount factor and  $\lambda$  is the GAE decay parameter. This formulation allows us to correct for the distributional shift between the policy used to generate the offline data and the current policy being optimized. We then perform the policy updates using policy gradient. We do not update the value function.

### 3.6. Evaluation

We evaluate all methods under two settings: training on optimal data (0.0 corruption) and partially corrupted data (0.5 corruption). For each setting, we report the final answer accuracy on a held-out test set.

Under higher corruption, **SFT** tends to suffer as it directly imitates the dataset; incorrect demonstrations lead to reduced performance. In contrast, **standard PPO** benefits from exploration and reward feedback, which can help the policy recover from poor supervision and improve accuracy over time.

**Hybrid PPO** begins training close to the reference policy. During training, since we are evaluating our policy on the sub-optimal dataset it selectively diverges from suboptimal parts of the reference policy as it gathers high-reward online rollouts. This allows Hybrid PPO to retain beneficial behaviors from the reference policy while improving upon them, offering a form of adaptive penalty unlike a fixed KL penalty.

## 4. Results

### SFT

As expected, increasing data corruption leads to a decline in model accuracy on the test dataset. Figure 1 shows the evaluation accuracy under 50% corruption (maximum accuracy  $\sim 17\%$ ) and under 0% corruption (maximum accuracy  $\sim 25\%$ ). The degradation confirms that supervised fine-tuning is sensitive to suboptimal demonstrations, as it directly imitates the corrupted dataset.

### ReFT (Standard PPO)

Building on the SFT baselines, we applied standard PPO to fine-tune the policy using reinforcement learning. However, due to memory constraints due to an extra value head, and suspected memory leaks, training runs on a Tesla V100 GPU failed silently—even at small batch sizes (e.g., 1). While full training runs could not be completed, we were able to evaluate the model after one PPO epoch. The resulting accuracy was approximately 19% for 0% corruption and 16% for 50% corruption. This initial drop in accuracy is likely due to the uncalibrated value head introducing instability in early policy updates. We expect that continued training would have led to improved performance.

### Hybrid PPO

Hybrid PPO further increases memory requirements, as it maintains two separate rollout buffers: one for offline data and one for online rollouts. These memory demands caused training to fail even at batch size 1. While complete runs were not successful, we monitored the log-probability of the gold CoT tokens under the current policy (Figure 2). Initially, these log-probabilities were high, but they declined as training progressed. This trend may reflect that the policy

initially being close to the reference policy and then slowly diverging. This could also be an artefact of using batch size as 1 and should not be interpreted conclusively. Due to the inability to complete full training runs, we were not able to obtain statistically meaningful evaluations for the Hybrid PPO algorithm.

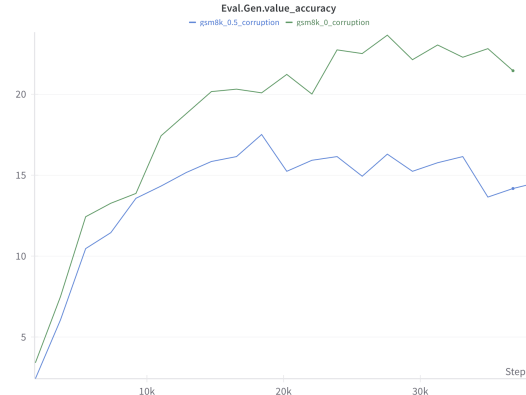


Figure 1. Answer accuracy over training steps for SFT trained on an uncorrupted dataset (ratio = 0) and a partially corrupted dataset (ratio = 0.5).

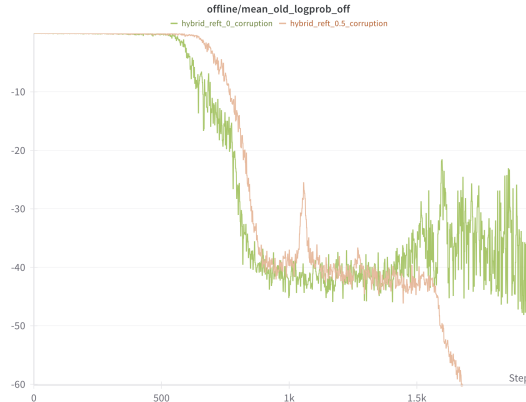


Figure 2. Gold CoT Log Probs through training for Hybrid PPO

## 5. Conclusion

In this work, we explored the impact of sub-optimal supervision on LLM fine-tuning using structurally corrupted versions of the GSM8K dataset. Our results demonstrate that SFT degrades significantly when trained using noisy sub-optimal offline datasets, while PPO can partially recover performance. Although GPU constraints prevented us from fully training PPO and hybrid PPO models, early signs indicate their potential to overcome sub-optimal initializations. While the findings are not conclusive, there remains promise of hybrid RL models for improving LLM robust-

ness in real-world settings where perfect supervision (i.e, optimal offline datasets) is rare.

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